
MUSCLE ACTIVITY AND FATIGUE

Brain Machine Interfaces



ABSTRACT

Learning how to use Vernier's hand dynamometer and EMG sensor. Understanding the signals obtained.

Carolina Martín Frisuelos -

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1. Background for the Experiment

In this laboratory practice a hand dynamometer and an ECG/EMG sensor were used. The hand dynamometer allowed us to record grip strength, using a sensor that converts the applied force into a voltage. The voltage of this device was recorded, and then using MATLAB, the force was computed back again using the linear relationship described by the script (Calibration Data: Force(N) = 175.416 *V -19.296).

Furthermore, three electrodes connected to a student's forearm were used to read the EMG electrical signal and measure the muscle's electrical activity. This sensor will measure the sum pf the action potentials due muscle activity.

Both Vernier's sensors (hand dynamometer and ECG/EMG sensors) were connected to a NI ELVIS II Series using an Analog Proto Board Connector.

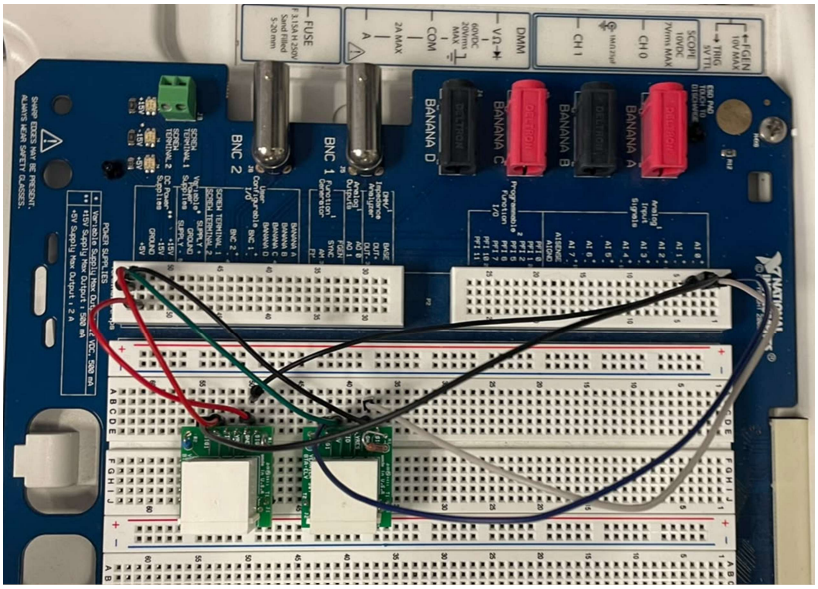


Figure 1. Vernier sensor attached to NI ELVIS through Analog Proto Board Connector

1.1 Differences between 'Force vs Time' and 'EMG vs Time' Signals

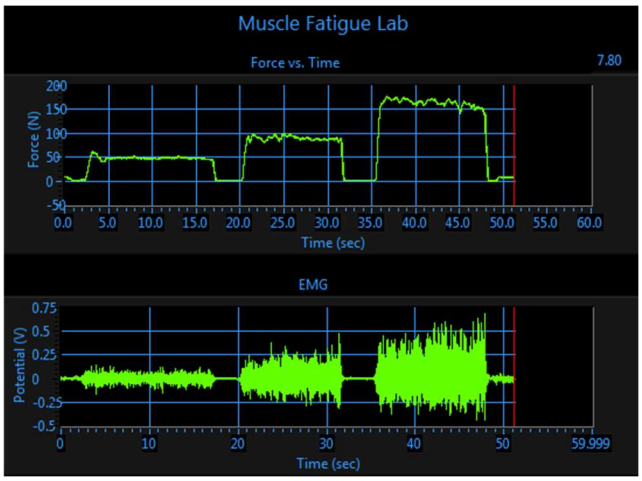


Figure 2. Comparison Script Figure (Force and EMG vs Time)

The signals of Force and EMG are two different physical quantities. The Force signal is obtained using strain-gage based technology, and it is a smoother and lower frequency measurement, in which there is a more proportional relationship between the voltage and the force used, thus when a student apply pressure to the device the voltage will go up. This voltage will then be transformed into Newtons using the device calibration linear relationship. On the other hand, the EMG signal comes from three electrodes placed on the forearm of the student, and it will pick up the action potentials coming from the muscle fibers. The relationship with EMG and the force applied by the student should also be proportional (more force implies more fibers recruited or more firing, thus higher and denser EMG signal), however, its relationship is more tendency-based, rather than a clean signal going up or down depending on force.

2. Grip Strength and Muscle Activity

To answer the data analysis proposed questions, let us remember that in this first experiment, we tested the grip strength of the students using the hand dynamometer and the EMG sensor in which we followed the Table 1's time and grip pattern, in which we also observe the tendencies that we should observe in the signals.

Table 1. Grip and Muscle Activity Experiment

Time	Grip / Force	EMG Band
0-20s	Normal grip (~30-70N)	Smaller and shorter
25-45s	50% grip (~40-100N)	Denser and Taller
50 – 60s	Maximum Grip (~40-100N)	Very Dense and Tall
In between rest seconds	No grip (~0N)	Should be almost no signal

Table 2. Maximum, Minimum, and Median Force for Each Student

Student Name	Maximum Force (N)	Minimum Force (N)	Median Force (N)
(1) Carolina Martin	174.51 N	-2.08 N	37.51 N
(2) Sandra Rodriguez	171.74 N	-0.73 N	12.83 N

1) **Analyze the screenshot of your personal data. Was there any noticeable correlation between muscle activity and grip strength?**

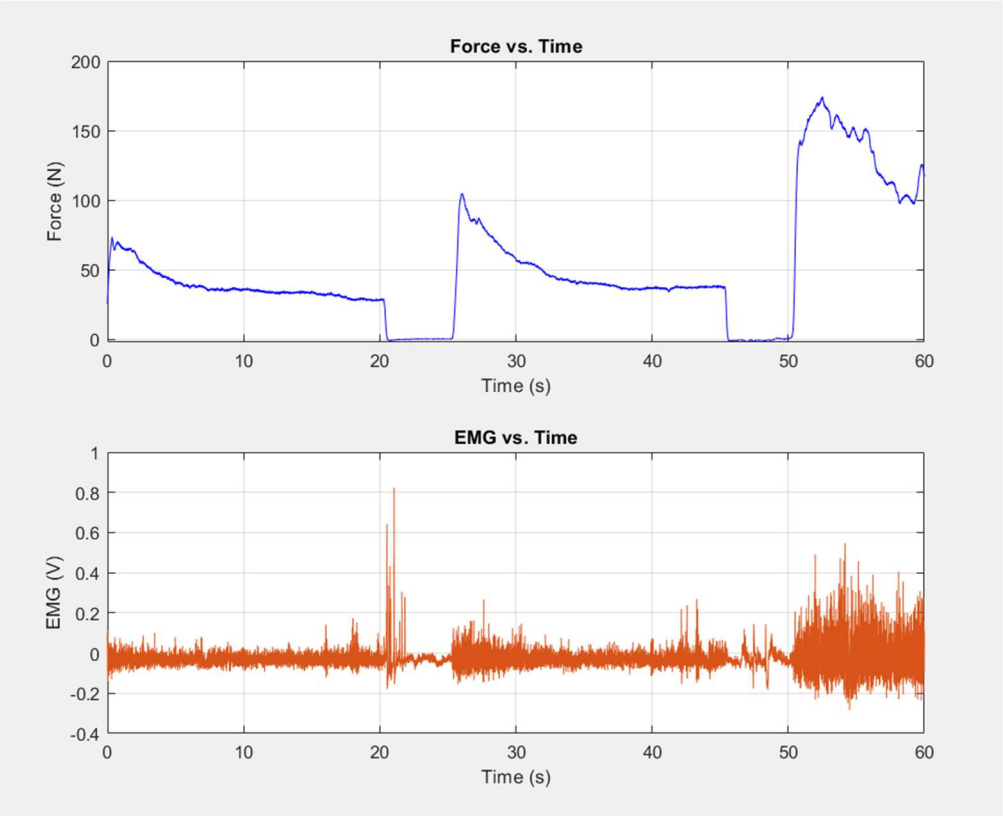


Figure 3. Force and EMG vs Time - student 1

Looking at the signals obtained from student 1 there is quite a relationship between applying more force and a higher voltage in the EMG signal. We can compare the first section when the student is applying a normal grip to the last section of maximum grip. There is a thinner EMG signal in the first case (moves approximately between -0.1V and 0.1V) and a higher amplitude signal in the last part (moves approximately between -0.2V and 0.2/0.4V). This tendency confirms the theory that when more force is needed the nervous system recruits more motor units and makes them fire faster. However, there are often errors or noise specifically in the EMG signal due to the movement of the electrodes or cables; or other muscle activity that is not involved in the grip strength, such as the stability of the arm or the position. In Figure 2, we can see one clear example, which would be the spike that occurs at around 21 seconds, in which the force is 0N and the EMG signal should be minimal, but it is not. Therefore, the spike is probably because of a mechanical or cable disturbance, not a muscle event.

2) **Compare the screenshot of your personal data with the other members in your group. Does their data appear to support the same trends you recorded in the previous question?**

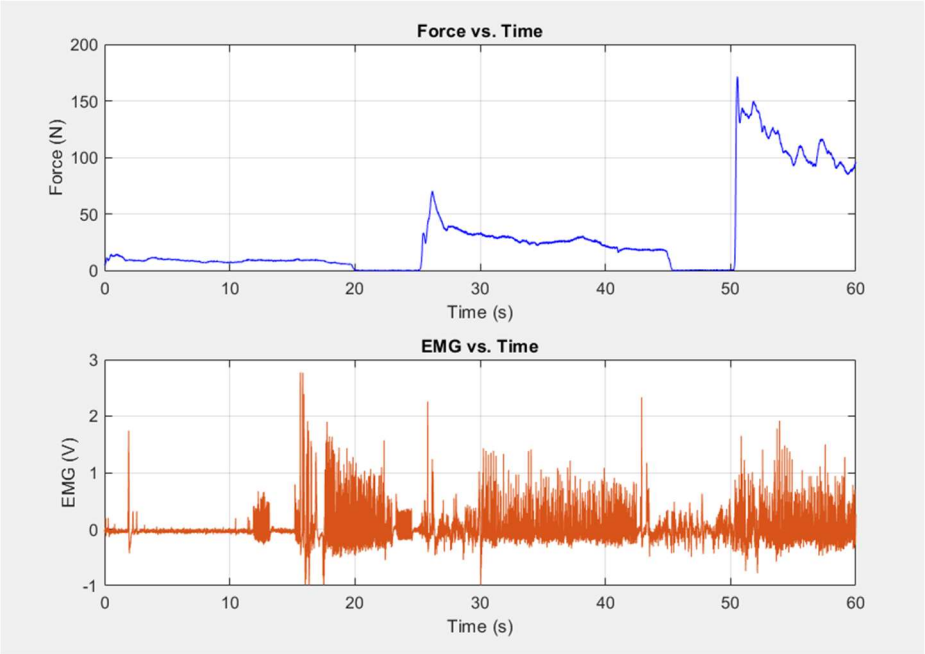


Figure 4. Force and EMG vs Time – student 2

For student 2 at the beginning when there is normal grip strength, we can see that the EMG is smaller for most of the first 10 seconds, which confirms again the tendency of proportional relationship between grip strength and the EMG signal. However, the rest of the EMG acquisition of student 2 is not very accurate. We can state this, because the voltage is reaching 1/2V when oscillating; while in student 1 and Figure 1 from the laboratory script the voltage stays lower than 1V. This indicates that there must be some cable or electrode distortion creating this; also, the student might have not been maintaining a correct posture or was moving the arm, even if it was not applying force in the hand dynamometer. This will all create a higher EMG signal, even when no grip strength was being recorded.

3) Open the saved files with MATLAB and obtain the correlation between the EMG amplitude and the exerted force. To do so, first rectify and obtain the envelop or a Low Pass Filtered version of the EMG signal.

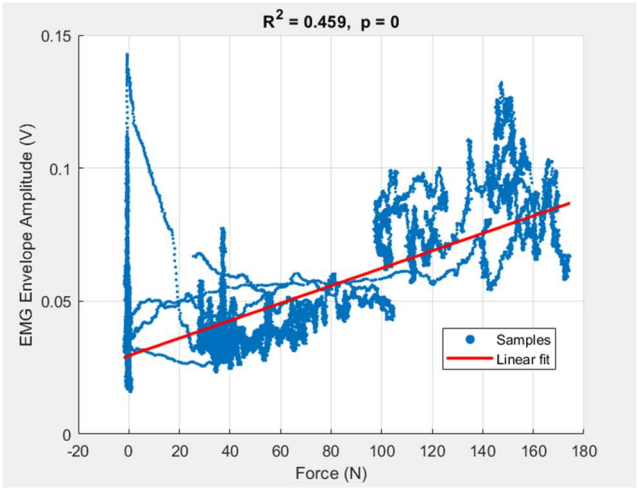


Figure 5. EMG amplitude and Exerted Force Correlation - student 1

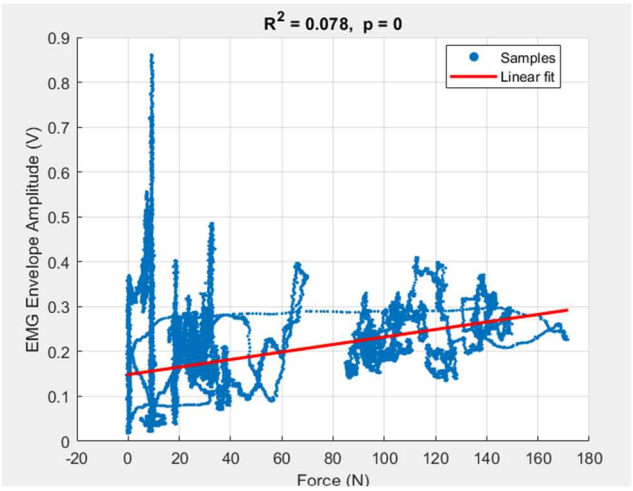


Figure 6. EMG amplitude and Exerted Force Correlation - student 2

4) **Show the correlation in a graph (and ideally, the fitted line as well). Display the R^2 value and the corresponding p value. Use adequate axis labels and units**

We can clearly see the red fit line going upward in both students, which demonstrates a positive correlation of EMG amplitude and the Exerted Force on the hand dynamometer. This trend is also bigger in student 1, since student 2 EMG signal was a bit distorted. Student 1 has a more condensed cloud of points sitting between 0.02-0.14V, which increases when more force is applied. Student 2 has a much larger y-axis and its voltage ranges between 0-0.9V and has a higher concentration of points up to ~0.85V when the force is minimal (around 0-20N). This demonstrates dominance in student 2 data of noise or artifact over muscle signal. It is important to mention that the first near-vertical streak on the left of both relationships may be also due to the rest transitions in which the EMG takes an instant more to adjust to the force going to almost 0 and decay from the level it was. This mismatch creates the artifact we observe in both plots.

Moreover, the R^2 is the fraction of the variation in the EMG which can be correlated or explained by the force applied. For a noisy biological data such as an EMG, student 1 R^2=0.459 indicates a real relationship between both measurements, while student 2 R^2=0.078 indicates almost no relationship and a data-acquisition issue. Finally, $p \approx 0$ in both is due the fact that there are thousands of samples obtained, so it states that the results are not because of chance, however, it cannot state any relationship with strength, which is better explained by the R and the linear fit.

3. Muscle Fatigue

3.1 Data Acquisition

Firstly, let us consider the conditions of each experiment in this section.

- **Part 2a (110s recording):** 20% strength (20 seconds) – Rest (10 seconds) – 40% strength (20 seconds) – Rest (10 seconds) – 60% strength (20 seconds) – Rest (10 seconds) – 80% strength (20 seconds)
- **Part 2b (140s recording, done twice — once per weight (light and heavy**)): hold at 90-degree weight (40 seconds) – Rest (10 seconds) – hold at 90-degree weight (40 seconds) – Rest (10 seconds) – hold at 90-degree weight (40 seconds)
- **Part 2c: Dominant vs Non-dominant Hand (35 seconds each hand):** 50% maximal grip strength (15 seconds) – Rest (10 seconds) – Maximal grip strength (5 seconds)

3.1.1. Student 1. Data

2a. Muscle Fatigue Over Time

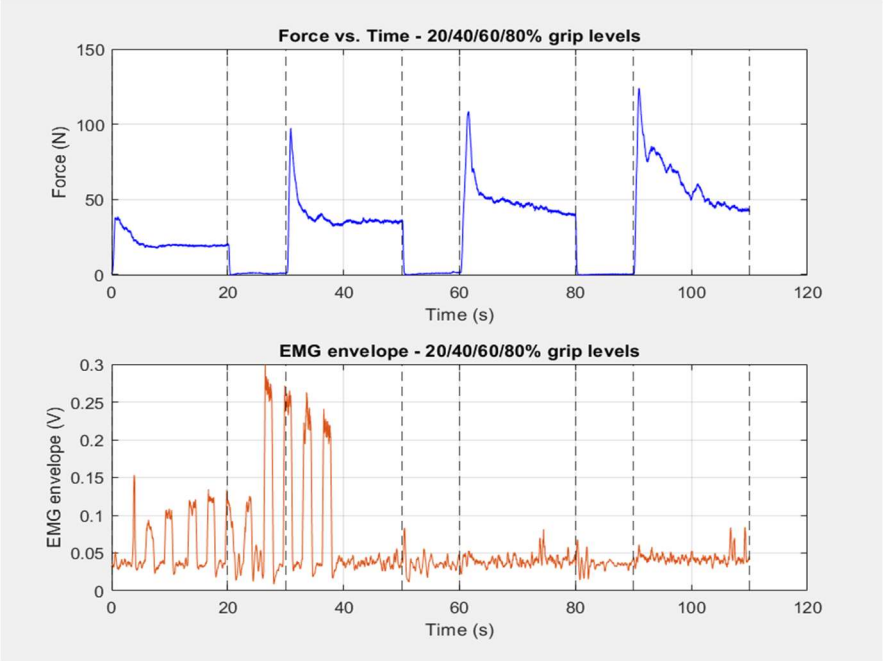


Figure 7. 2a Student 1-Muscle Fatigue-Grip Percentages- Force and EMG Signals

Table 3. 2a Student 1-Muscle Fatigue-Grip Percentages- Maximum Force and Mean EMG

Grip Level	Max Force (N)	Mean EMG (V)
20%	38.47	0.0556
40%	97.42	0.0742
60%	108.60	0.0402
80%	124.08	0.0433

2b. Muscle Fatigue Over Time

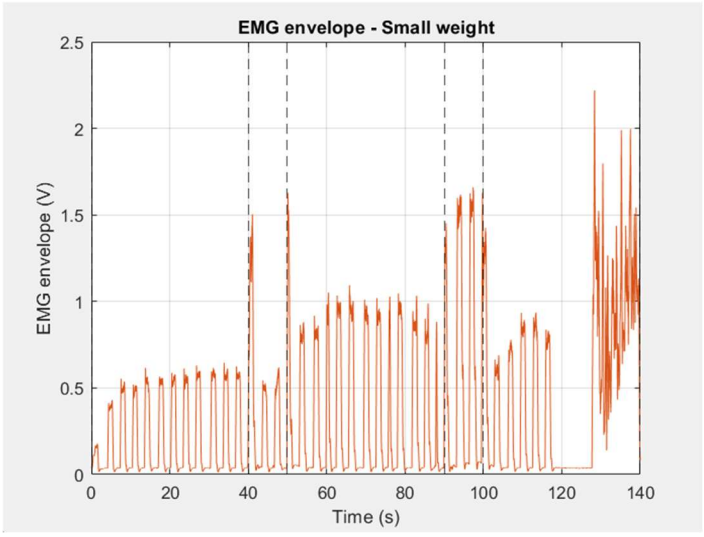


Figure 8. 2b Student 1- Muscle Fatigue - EMG small weight

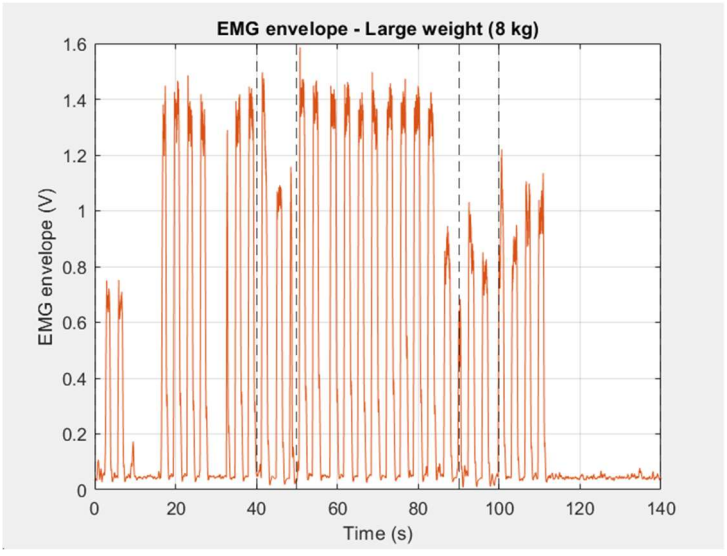


Figure 9. 2b Student 1- Muscle Fatigue - EMG large weight

Table 4. 2b. Student 1 - Holding weight Values

Hold	Time Window	Small Weight – Mean EMG (V)	Large Weight (8kg) – Mean EMG
1	0s-40s	0.2040	0.3476
2	50s-90s	0.3343	0.5791
3	100s-140s	0.2635	0.1647

2c. Dominant vs Non-dominant Hand

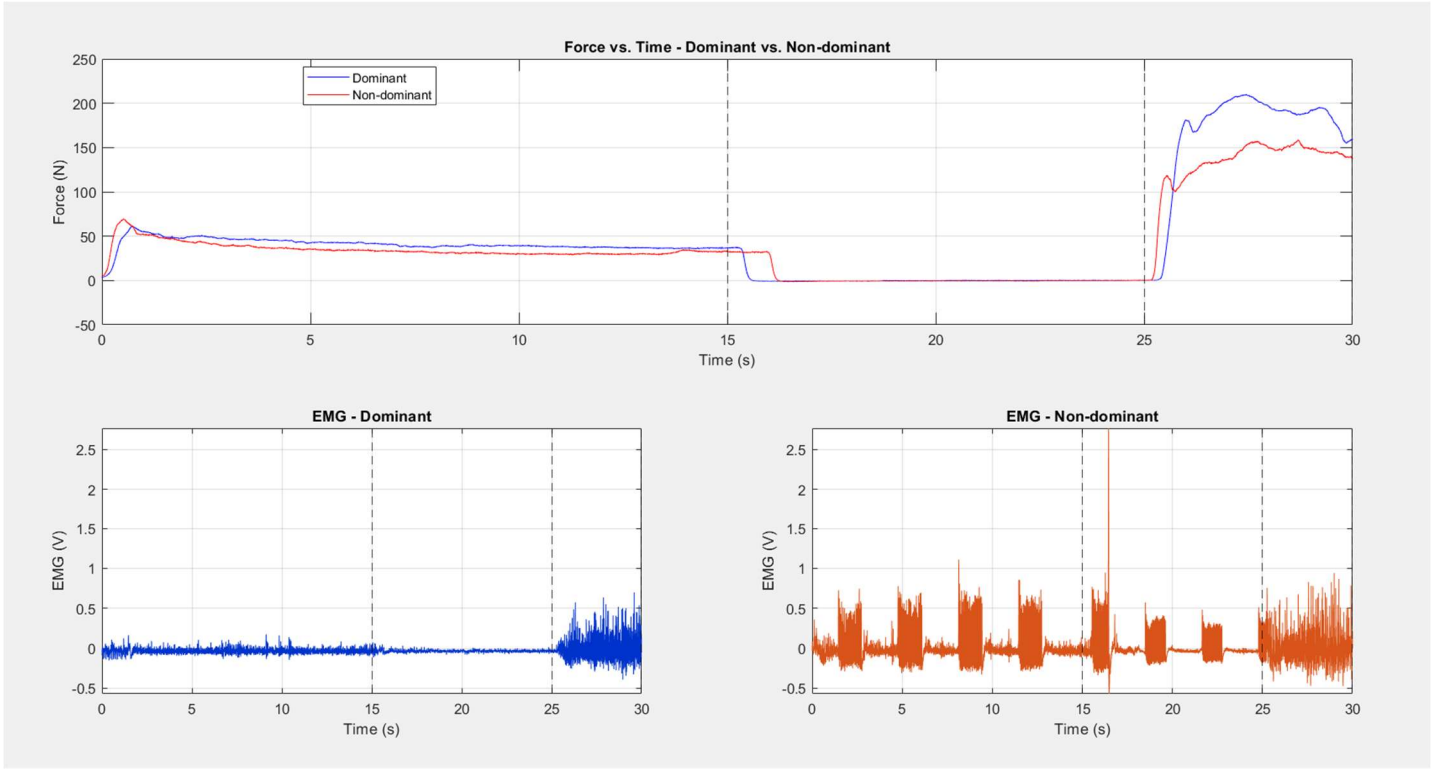


Figure 10. Student 1- Dominant vs Non-dominant Hand

3.1.2. Student 2. Data

2a. Muscle Fatigue Over Time

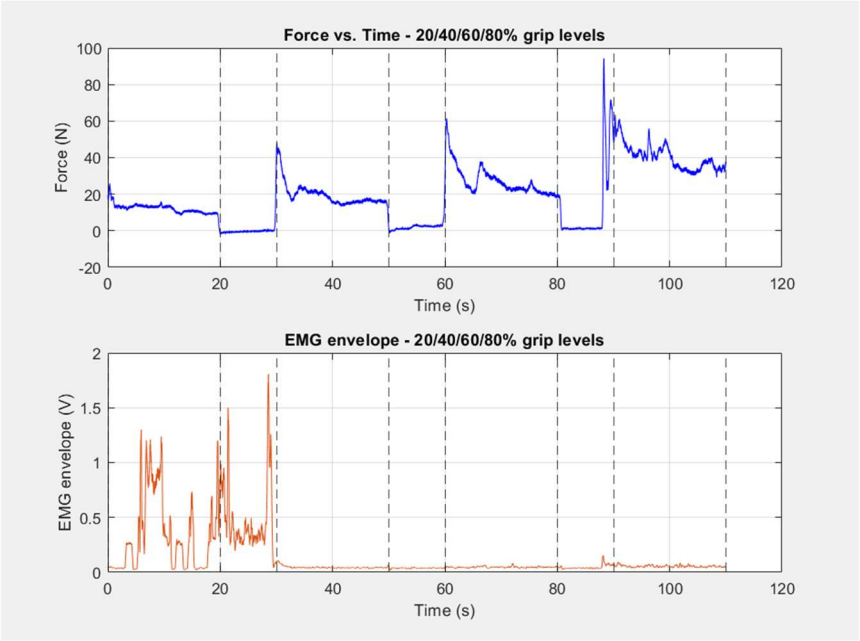


Figure 11. 2a Student 2-Muscle Fatigue-Grip Percentages- Force and EMG Signals

Table 5. 2a Student 2-Muscle Fatigue-Grip Percentages- Maximum Force and Mean EMG

Grip Level	Max Force (N)	Mean EMG (V)
20%	25.99	0.3317
40%	48.12	0.0456
60%	61.45	0.0469
80%	63.43	0.0546

2b. Muscle Fatigue Over Time

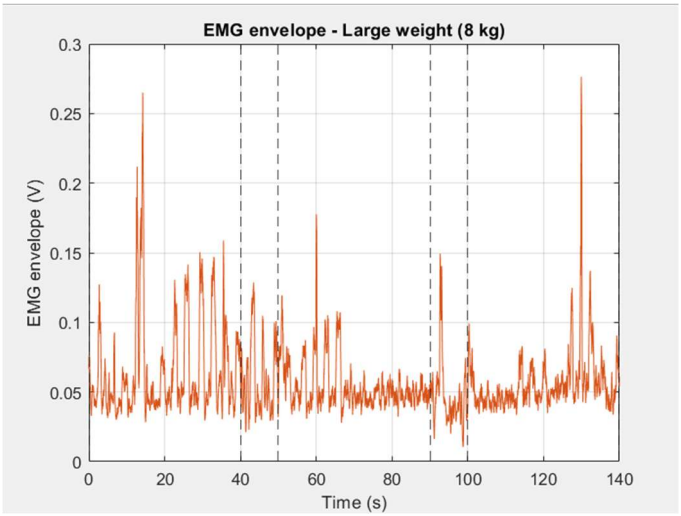


Figure 12. 2b Student 1- Muscle Fatigue - EMG small weight

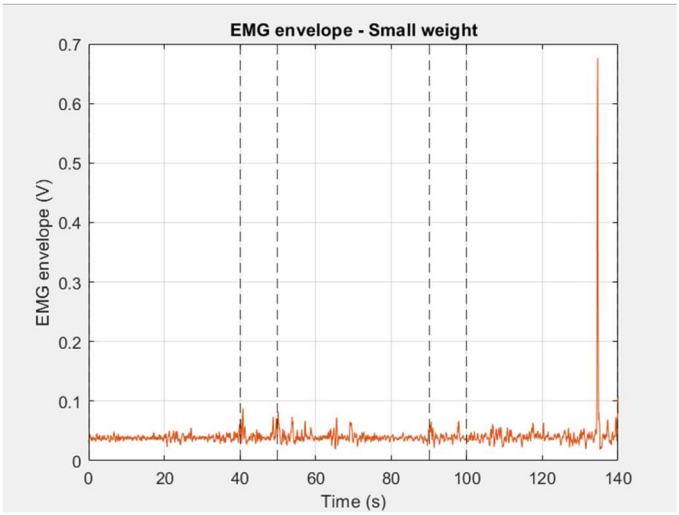


Figure 13.2b Student 2- Muscle Fatigue - EMG large weight

Table 6. 2b. Student 2 - Holding weight Values

Hold	Time Window	Small Weight – Mean EMG (V)	Large Weight (8kg) – Mean EMG
1	0s-40s	0.0386	0.0666
2	50s-90s	0.0397	0.0546
3	100s-140s	0.0445	0.0565

Data Analysis

Table 7: Maximum Grip Strength

Student Name	Maximum Force for 20% (N)	Maximum Force for 40% (N)	Maximum Force for 60% (N)	Maximum Force for 80% (N)
(1) Carolina Martin	38.47	97.42	108.60	124.08
(2) Sandra Rodriguez	25.99	48.12	61.45	63.43

1) **Analyze the screenshot of your personal data.**

- **Was there a noticeable trend or difference for 20%, 40%, 60%, or 80% maximal grip strength?**

There is a noticeable trend in the Force signals which obviously increases with the percentage of the strength, as we can see in the Maximum Grip Strength Table. Moreover, the EMG signals should increase with the grip strength percentage used, however, looking at both students EMG signal mean, you cannot really tell any trend in this part of the experiment.

- **Were you able to maintain 20% for the entire 20 seconds? What about 40%, 60%, and 80%?**

In both students, the 20% grip was maintained for almost the entire 20 seconds, as we can see in the Force vs Time plots. Nevertheless, in the rest of the percentages (40%, 60%, and 80%) for both students there is one initial peak and then the force recorded starts reducing for the first seconds and then stays quite stable for the final seconds of the time slot.

- **Why do you think you were able/not able to maintain the same grip strength for each percentage?**

I believe it is a low perception of one self's grip strength, in addition to the fatigue of the muscle fibers. This can biologically be explained by the type of fibers being used. The first ones to be recruited (used in the lower grip percentage) are more resistant to fatigue. Then, when you need more force to maintain the grip, the fast fibers (medium and large) will be recruited because they

can produce more force, but they fatigue quicker. This may be why the students were not able to maintain the higher percentages for a long time.

1) **and 2) What happened for the longer you maintained the weight? / Compare your data with the data of the other members in your group. Does their data confirm the trends you found in your own data?**

Theoretically, maintaining a longer time the weight will increase the amplitude of the EMG, which means the nervous system will be recruiting more fibers to maintain the corresponding weight in a 90-degree position. In our case it is hard to differentiate the trend in the EMG signals of the students for the experiment 2b) of the holding weights. However, if we compute the EMG Mean signal (look at data acquisition of 2b. for both students), when analyzing the small weight for both students the signal increases for each of the three 40 second periods when the weight needs to be held. So, each time the student picks up the weight again to hold it, a higher EMG signal mean value is observed. For the large weight, there is a less visible trend in the EMG mean values. Since the last values often drop, this could be an error, and also the summation that in the hold 3 there is a lot of fatigue that there cannot be more signals recruited if that is possible. Moreover, in the resting periods we observe often large signals probably because of the movement of the arm to drop the weight.

2) **OPTIONAL: Did gender appear to make a difference in your data? What about fitness level?**

We could not obtain the data between a male or a female. We obtained the data of two females. However, student 1 should have better grip strength than student 2 because of the involvement in exercise activities that use the grip such as strength training or climbing. We can observe a certain tendency in the Force signals of student 1 to have a higher grip strength, however, it is difficult to compare the EMG signals, since there has been a lot of distortion of the cables or other muscle activity in both students.

3) **Quantitatively and qualitatively describe how each of the 4 varying factors affected muscle fatigue.**

Factor	**Quantitative**	**Qualitative**
1. Grip Intensity (%) (20→40→60→80%)	- Student 1 Force (N): 38→97→109→124 N - Student 2 Force (N): 26→48→61→63 N	The higher the intensity = the more motor units recruited = more forced used = faster fatigue (student could not maintain the force) = decays in the higher percentages.
2. Weight Size	- Student 1 EMG mean value (first 40s): 0.2040 V (small weight) / 0.3476V (large weight) - Student 2 EMG mean value (first 40s): 0.0386 V (small weight) / 0.0666V (large weight)	A heavier load demands more motor units from the beginning of the contraction = higher EMG signal
3. Duration/ Repetition	- Student 1: 0.204→0.334→0.464 V (small weight) / 0.348→0.579→0.165 V (large weight)	Repeated effort increases fatigue; which increases the demand of motor units, but with enough cumulative demand (large weight),

	- Student 2: 0.0386→0.0397→0.0445 V (small weight) / 0.0666→0.0546→0.0565V (large weight)	compensation runs out entirely by Hold 3 (or it is an error of the signal when dropping the weight)
4. Hand Dominance	- 50% grip Force: dominant is ~2-3N higher than non-dominant hand - Maximal grip Force: dominant is ~50N higher than non-dominant hand - Non-dominant has higher EMG amplitude.	Dominant hand sustains higher force and needs less EMG per Newton than non-dominant.

4) Does the data from each member in the group confirm similar trends?

As mentioned before, the data in both students confirm the fact that more percentage of grip strength will require more Force, which can be seen in the plot and the maximum Force of both student in the experiment 2a), and also the increase in the mean value of EMG, when more grip strength percentage is needed. In 2b) a trend in the small weight of a higher EMG mean value was observed when more time has passed holding the weight. But it was not repeated for the large weight.

5) Which factor had the biggest effect on muscle fatigue? Why do you think this is?

The factor that had the biggest effect on muscle fatigue was probably the large weight. As we can observe in the EMG mean value of student 1, since for the first two holds in the EMG rose from 0.348 to 0.579V, but on hold 3 it crashed to 0.165V. Which can indicate that the nervous system runed out of motor units to recruit and cannot hold anymore the weight.

6) Did you notice any trends in the EMG graphs for each of the varying factors? What does this say about the electrical activity in your muscles?

1. Grip Intensity: The EMG generally scaled with force, which will mean more neural drive to increase the force.
2. Weight Size: Before muscle fatigue, in both students the EMG had higher mean values, the bigger the weight size, thus a heavier load means more motor units recruited.
3. Duration/Repetition: When looking at the three different holds of the weights, the EMG increased its value with time, until muscle fatigue and no more muscle fibers to recruit, EMG signal collapses.
4. Hand dominance: Non-dominant hand had a higher EMG signal overall (bigger and denser), despite having lower force at the maximum grip. This shows that the dominant hand can produce more force efficiently, by needing to recruit less motor units.

7) **Do you think electrode placement had any effect on your data? Can you think of a way to increase the accuracy of your data by changing the number and/or placements of the electrodes?**

Electrode placement may be one of the reasons why in the first experiments Student 1 and 2 EMG plots were at different voltage scales for the same task. Moreover, their correlation between strength and EMG ($R^2 = 0.459$ vs. 0.078) had a huge difference, which can also imply a different placement of the electrodes.

To increase the accuracy, we should make sure that the electrodes are placed close to the forearm flexor muscles and not the tendon or bone area. Also, keep the same more accurate distance between electrodes; or even clean the skin before placing the electrodes. One of the things we could improve the most is to secure the cable at one place so that it does not create sharp artifacts like most of the spikes we can see in the EMG signals. Finally, we could add a second electrode pair at a different site and compare the two EMG signals, to check if the spikes or bursts appear in both or if it is just a local artifact not produced by muscle signal.

8) **Are there any sources of error that might have contributed to any unexpected trends?**

As mentioned all through the report, some of the sources of error that may have contributed to unexpected trends are:

- Motion or cable artifacts (Mechanical Disturbance): have probably created sharp and sudden spikes when they did not make sense, such as in resting times or when dropping the weight.
- No knowledge in the real force being applied: students had to guess the grip they were applying, and they may not have known the real percentage of their grip strength.
- Electrode drifting: loosening through the experiment and using the same electrodes for Student 1 and 2, may have also decreased the quality of our EMG signals.
- Lack of procedure: Students may have been standing up or sitting during the experiment, also the 90 degree position hold of the arm was not carefully measure, and there was no stated way to rest the arm in the resting time, so that may have created different artifact or different fatigue recovery.